Measuring the Multilingual Token Tax: Frontier Premiums and a Language-Conditional SuperBPE Bake-Off

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Abstract

Production language-model tokenizers remain English-centric: the same parallel content can require substantially more tokens in under-served scripts, inflating latency and—because commercial APIs bill per token—inference spend [Petrov et al., 2023, Ahia et al., 2023]. We report three complementary studies. First, we publish an empirical *token-tax* table on FLORES-200 for Amharic, Odia, and dialectal Arabic (Egyptian, Moroccan), with Swahili and Hausa as Latin-script African controls, across frontier BPE tokenizers and a published SuperBPE baseline. On OpenAI `o200k`, Amharic and Odia require 5.78× and 4.99× as many tokens as English for the same sentences; Llama and GLM push Odia above 12×. These measured premiums replace the literature's broad 3–12× range with language- and tokenizer-specific numbers. Second, we train matched BPE and SuperBPE tokenizers [Liu et al., 2025] on an equal-byte six-language corpus (960 MB) and evaluate English-relative token premium, fertility, and Zipf deviation on FLORES. SuperBPE lowers fertility for every whitespace-delimited language in the set, yet *raises* English-relative premium for Mandarin (0.912 → 1.082) and Hungarian (1.084 → 1.160)—failure modes consistent with prior BPB evidence that SuperBPE underperforms BPE on these languages [Chowdhury and Woolf, 2026], and that make any tokenizer recommendation language-conditional. Third, we run a Phase 0 language-model pilot: matched `olmo2_100M` models trained for ∼2B equal tokens per arm. Raw bits/token favors BPE by +0.84 micro-average, but fertility-normalized approximate bits-per-byte collapses the gap to roughly +0.04 on non-CJK languages, with SuperBPE slightly better only on Haitian Creole. A planned 1B-parameter UniMax bake-off with true UTF-8 bits-per-byte evaluation is specified; those tables are left empty pending training.

1 Introduction

Tokenization converts Unicode text into a finite vocabulary of subword (or superword) units. Although the map is language-agnostic in principle, vocabularies and merge schedules of production systems are overwhelmingly tuned to English and other high-resource Latin-script languages [Petrov et al., 2023, Arnett et al., 2025]. The result is systematic over-segmentation of other languages: longer token sequences for the same meaning, shorter usable context windows, and—crucially for deployed systems—higher API bills, because vendors charge per token [Ahia et al., 2023].

Prior work has framed this disparity as a "token tax" or "script tax," often summarizing the gap with wide ranges such as 3–12× English [Dixit and Dixit, 2026, Somide, 2026, Petrov et al., 2023]. Such ranges are useful as existence proofs but are hard to use for planning: they mix tokenizers, language samples, and metrics. We contribute a locked, reproducible measurement on FLORES-200 [Goyal et al., 2022, NLLB Team et al., 2022] for a geographically diverse 12-language set, with a highlighted *target-script* slice—Amharic (Ethiopic), Odia (Orya), Egyptian and Moroccan Arabic—and Latin-script African controls (Swahili, Hausa).

A second question is whether algorithm choice can close the tax. SuperBPE [Liu et al., 2025] extends BPE with a second stage that learns whitespace-bridging "superwords," reporting large English-centric efficiency and quality gains. We ask whether those gains transfer under a multilingual equal-byte training regime that includes languages where SuperBPE has already been shown to underperform on BPB—Mandarin and Hungarian [Chowdhury and Woolf, 2026]. We train matched BPE and SuperBPE arms on 960 MB of FineWeb /FineWeb-2 text [Penedo et al., 2024, 2025] and evaluate on FLORES. We find that SuperBPE is not uniformly preferable: fertility improves for five of six languages, but English-relative token premium worsens for Mandarin and Hungarian, aligning with that prior extrinsic result. Any recommendation must therefore be language-conditional.

Downstream language-model quality is the natural next measurement. In §6 we report a Phase 0 `olmo2_100M` pilot under equal-token training and specify a deferred 1B UniMax bake-off [Chung et al., 2023] whose bits-per-byte tables remain empty pending training.

Contributions.

1. Encode-only FLORES token-tax audit for Amharic, Odia, dialectal Arabic, plus Swahili/Hausa controls; full matrices in the appendix.
2. Matched SuperBPE-versus-BPE bake-off on six languages at equal-byte scale, with Mandarin and Hungarian as SuperBPE premium failures.
3. A Phase 0 100M LM pilot (bits/token and fertility-normalized approx. BPB) plus a deferred 1B UniMax BPB evaluation skeleton.

2 Related Work

Cross-lingual tokenizer unfairness. Petrov et al. [2023] show that parallel text can differ by up to roughly 15× in token length across languages under multilingual tokenizers, and argue for fairer vocabularies. Ahia et al. [2023] connect the same phenomenon to commercial API pricing. Subsequent "script tax" and "African language tax" studies [Dixit and Dixit, 2026, Somide, 2026] report fertility and English-relative premiums on African and Indic scripts. Arnett et al. [2025] operationalize *token premiums* as corpus token-count ratios on FLORES; morphology-focused analyses [Arnett and Bergen, 2024] motivate including agglutinative languages such as Hungarian.

Superword tokenization. Liu et al. [2025] introduce SuperBPE: stage 1 learns ordinary subwords under whitespace pretokenization; stage 2 lifts that restriction to form superwords. Published SuperBPE models improve English-centric compression and 8B downstream scores. Chowdhury and Woolf [2026] provide the first extrinsic cross-language BPB comparison of BPE, SuperBPE, and MorphBPE on English, Mandarin, and Hungarian: SuperBPE matches BPE on English but underperforms by 0.01–0.06 BPB on Hungarian and Mandarin, suggesting that cross-whitespace merging is counterproductive outside English. Our Study B tests whether encode-time premiums under equal-byte multilingual training reproduce that language-conditional pattern.

Multilingual pretraining mixtures. Chung et al. [2023] propose UniMax sampling with an epoch cap to balance languages without over-repeating scarce data. Our Study C 1B bake-off (§6) adopts UniMax over FineWeb-2 pools with a 20B-token budget measured on the BPE arm [Hoffmann et al., 2022, Muennighoff et al., 2023, OLMo Team et al., 2025].

3 Metrics

Let $\mathbf{s}_{1:N}^{(\ell)}$ be the N parallel FLORES-200 `devtest` sentences in language ℓ (N=1012). For a tokenizer h we define corpus token count

$$\mathrm{CTC}(\ell; h) \;=\; \sum_{i=1}^{N} \left| \, h\big(\mathbf{s}_i^{(\ell)}\big) \, \right|, \tag{1}$$

excluding BOS/EOS where the tokenizer would insert them. Following Arnett et al. [2025], the **token premium** relative to English is

$$\mathrm{premium}(\ell; h) \;=\; \frac{\mathrm{CTC}(\ell; h)}{\mathrm{CTC}(\mathrm{eng}; h)}. \tag{2}$$

By construction, premium(eng; h) = 1 for every h. **Fertility** is tokens per whitespace word after Unicode NFKC [Petrov et al., 2023]; for Mandarin we also report tokens per character and caution against interpreting tokens/word. **Characters per token** uses non-whitespace code points over CTC.

API cost scaling. Commercial language-model APIs bill by token [Ahia et al., 2023]. For a fixed parallel payload, relative inference token spend therefore scales linearly with premium: if language ℓ has premium p, it incurs p times as many billed tokens as English under the same tokenizer and pricing tier. We report premiums rather than dollarizing a student cohort; the conversion to currency is a single multiplication by the vendor's per-token rate.

4 Study A: Frontier Token Tax

4.1 Setup

Languages. We evaluate the locked 12-language FLORES-200 set used by our efficiency validation: Swahili, Hausa, Amharic, Odia, Mandarin, Egyptian Arabic, Moroccan Arabic, Hungarian, Ukrainian, English, Ayacucho Quechua, and Guarani. The *target-script* highlight is Amharic (`amh_Ethi`), Odia (`ory_Orya`), Egyptian Arabic (`arz_Arab`), and Moroccan Arabic (`ary_Arab`), with Swahili (`swh_Latn`) and Hausa (`hau_Latn`) as Latin-script African controls.

Tokenizers. Frontier BPE: OpenAI `o200k_base` [OpenAI, 2024], Llama 3.1 [Meta AI, 2024], Qwen2.5 [Qwen Team, 2024], GLM-5.2 [Zhipu AI, 2025]. Published SuperBPE: UW OLMo2-8B SuperBPE (t180k) [University of Washington, 2025, Liu et al., 2025]. Multilingual baseline: NLLB-200 Unigram [NLLB Team et al., 2022]. All metrics are encode-only; no language-model generation is required.

Decision rule (pre-registered). We treat a fairer-tokenizer research program as worth pursuing if either (A) at least one non-English language has premium ≥ 2 on ≥ 2 frontier BPE tokenizers, or (B) mean fertility for a non-Latin / complex-morphology language is $\geq 2\times$ English on the same tokenizer. Both clauses pass: five languages hit clause A (`amh_Ethi`, `ory_Orya`, `quy_Latn`, `grn_Latn`, `hun_Latn`).

4.2 Results

Table 1 replaces the literature's undifferentiated 3–12× band with measured numbers. On `o200k`, Amharic (5.78×) and Odia (4.99×) sit in the middle-to-upper part of that band; Llama and GLM push Odia above 12×, matching the upper end of prior summaries. Dialectal Arabic

Table 1: English-relative token premium on FLORES-200 `devtest` for target scripts and African Latin-script controls. Values are $\mathrm{CTC}_\ell/\mathrm{CTC}_{\mathrm{eng}}$. Full 12-language matrix in Table 10.

Language	o200k	Llama	Qwen	GLM	SuperBPE
English	1.00	1.00	1.00	1.00	1.00
Swahili (control)	1.49	1.93	1.94	1.93	2.43
Hausa (control)	1.55	1.97	1.94	1.94	2.47
Egyptian Arabic	1.41	1.65	1.63	1.75	3.22
Moroccan Arabic	1.48	1.71	1.69	1.81	3.26
Amharic	**5.78**	7.59	4.05	7.59	5.90
Odia	**4.99**	**12.16**	9.70	**12.36**	13.53

Table 2: Continent mean token premium (non-English languages only).

Continent	o200k	Llama	Qwen	SuperBPE
Africa	2.94	3.83	2.64	3.60
Americas	1.80	2.08	2.07	2.66
Asia	2.28	4.20	3.51	5.55
Europe	1.77	1.88	2.31	3.38

is milder (1.4–1.8× on frontier BPE) but still a clear tax relative to English; the published SuperBPE baseline nearly doubles those Arabic premiums (3.2×). Swahili and Hausa—Latin script, African—land near 1.5× on `o200k` and just under 2× on Llama/Qwen/GLM, confirming that script family alone does not explain the Amharic/Odia gap.

Continent means (Table 2, Figure 1) show Africa and Asia carrying the heaviest mean tax under frontier BPEs; Asia's SuperBPE mean (5.55) is dominated by Odia. Because API cost scales with tokens, an Amharic request that is content-parallel to an English request costs 5.78× as many `o200k` tokens—and therefore 5.78× the token-metered spend—under the same pricing tier.

5 Study B: SuperBPE versus BPE

5.1 Setup

Languages (separate scope). Study B uses a different locked six-language set chosen for FineWeb-2 availability and typological coverage: English, Hungarian, Mandarin, Hindi, Swahili, and Haitian Creole (`eng_Latn`, `hun_Latn`, `zho_Hans`, `hin_Deva`, `swh_Latn`, `hat_Latn`). This set is *not* interchangeable with Study A's twelve languages; Amharic, Odia, Hausa, and dialectal Arabic are absent from the bake-off.

Training. We train matched BPE and SuperBPE tokenizers with Gigatoken [Rød, 2026] via our SuperBPE-capable fork `supergigatoken` [Verma, 2026] on an equal-byte *scale* corpus: 160 MB per language, 960 MB total, drawn from FineWeb (English) and FineWeb-2 (other languages). Equal bytes prevent a measured Creole or Swahili penalty from being a corpus-size artifact. Vocabulary size is 100,000; SuperBPE transitions at 80,000. Pair verification confirms that SuperBPE carries the BPE merge prefix through the transition point. We pin `supergigatoken` at commit `00e61db` (branch `superbpe-stage1-pretokenizer`). Each arm is encoded with the pre-tokenizer recorded in its own `tokenizer.json`.

Evaluation. FLORES-200 `devtest`: English-relative token premium, fertility (tokens/word), and Zipf–Mandelbrot Kolmogorov–Smirnov deviation ks_zipf [Clauset et al., 2009] under a

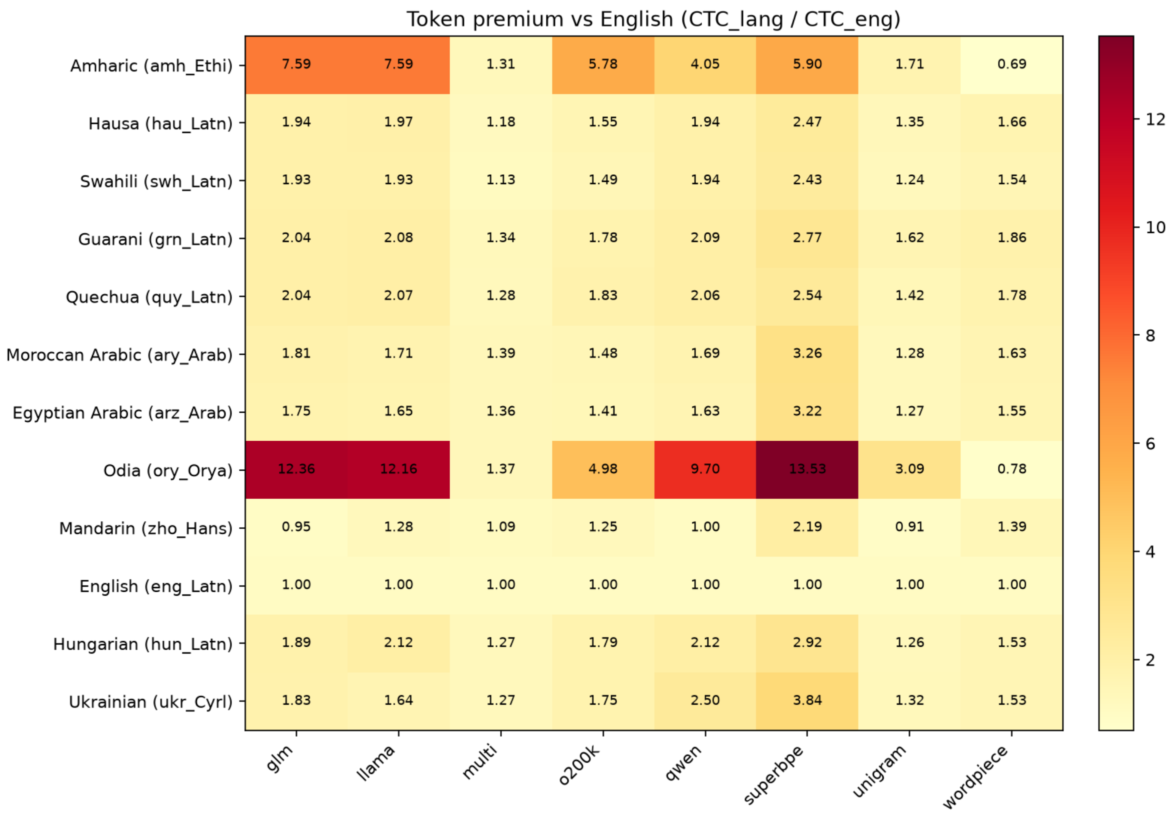

Figure 1: Token-premium heatmap over the full 12-language efficiency set (Study A). English is exactly 1.0 by construction.

matched-sentence, token-unit regime. Lower ks_zipf is closer to a pure Zipf distribution. LM bits-per-byte is deferred to Study C (§6).

5.2 Tokenizer-level results

Tables 3–5 and Figure 2 show a mixed picture. SuperBPE reduces fertility for every non-Mandarin language (Table 4), consistent with superwords packing multi-word expressions. Yet English-relative *premium*—the quantity that tracks parallel API cost—moves the wrong way for Hungarian and Mandarin (Table 3), the same pair where Chowdhury and Woolf [2026] found SuperBPE worse than BPE by 0.01–0.06 BPB. Mandarin's BPE premium below 1.0 (0.912) indicates that, under this equal-byte vocabulary, Mandarin is cheaper than English in tokens; SuperBPE erases that advantage (1.082). Hungarian's premium rises from 1.084 to 1.160. Haitian Creole is the clearest SuperBPE win on premium (1.034 → 1.004); Hindi is essentially tied. On Zipf structure (Table 5), BPE sits closer to a pure Zipf law for all six languages.

Language-conditional recommendation. Under equal-byte multilingual training and FLORES premium, we do *not* recommend SuperBPE as a blanket upgrade. Prefer SuperBPE where fertility and premium both improve (here: Haitian Creole; Hindi approximately neutral on premium with large fertility gain). Prefer BPE where premium worsens—explicitly Mandarin and Hungarian in this suite—unless a downstream LM metric overturns the encode-only ranking.

Table 3: Study B: English-relative token premium after equal-byte training. SuperBPE *worsens* premium for Hungarian and Mandarin (bold).

Language	BPE	SuperBPE	Δ
English	1.000	1.000	—
Hungarian	1.084	**1.160**	+0.076
Mandarin	0.912	**1.082**	+0.170
Hindi	1.259	1.256	−0.003
Swahili	1.032	1.056	+0.024
Haitian Creole	1.034	1.004	−0.030

Table 4: Study B: fertility (tokens / whitespace word). Mandarin is reported but not interpreted as tokens/word.

Language	BPE	SuperBPE
English	1.321	1.124
Hungarian	1.676	1.526
Mandarin	12.759	12.874
Hindi	1.420	1.206
Swahili	1.402	1.221
Haitian Creole	1.293	1.068

5.3 Trainer caveat (Hindi)

Our default trainer (`supergigatoken` [Verma, 2026], built on Gigatoken [Rød, 2026]) implements SuperBPE stage 2 without the official `STAGE2_REGEX`, bounding units by separator, newline, and a maximum unit length of 128. Relative to the official SuperBPE trainer [Liu et al., 2025], this yields a Hindi-specific compression advantage for the SuperBPE arm at small vocabularies (premium deltas of −3.13% at 4k and −6.61% at 16k in pilot cross-checks). The BPE arm remains comparable across trainers. Hindi SuperBPE premiums in Table 3 should therefore be read as *gigatoken-SuperBPE* numbers, not as a direct replication of published Liu et al. SuperBPE.

6 Study C: Language-Model Bake-Off

Tokenizer-level metrics do not settle whether SuperBPE's fertility gains translate into better language modeling under a fixed compute budget. Prior monolingual BPB evidence already favors BPE over SuperBPE for Mandarin and Hungarian [Chowdhury and Woolf, 2026]. Study C asks the same question under matched multilingual pretraining with the Study B Gigatoken / `supergigatoken` tokenizers [Rød, 2026, Verma, 2026]. We execute a cheap Phase 0 pilot first, then specify a Phase 1 $\sim$1B UniMax run whose result tables remain empty.

6.1 Experiment process

Both phases share the Study B tokenizers (`tokenizer/gigatoken-bpe/v1` and `tokenizer/gigatoken-superbp` vocabulary size 100,000, SuperBPE transition at 80,000) and the same six FineWeb /FineWeb-2 languages. Phase 0 materializes equal-*token* packed shards, trains matched `olmo2_100M` models, evaluates held-out cross-entropy as bits/token, then fertility-normalizes with Study B FLORES characters-per-token. Phase 1 (deferred) materializes UniMax equal-*byte* shards, trains matched `olmo2_1B` models to equal FLOPs, and will report true UTF-8 bits-per-byte.

Table 5: Study B: Zipf deviation ks_zipf (lower = closer to Zipf). BPE is closer for all six languages.

Language	BPE	SuperBPE
English	0.121	0.297
Hungarian	0.200	0.325
Mandarin	0.292	0.310
Hindi	0.134	0.301
Swahili	0.163	0.304
Haitian Creole	0.080	0.294

6.2 Phase 0: 100M equal-token pilot

Training specifications.

- **Model:** `olmo2_100M` via OLMo-core [OLMo Team et al., 2025].
- **Corpora:** `pretrain/fineweb2-phase0-equal-{bpe,superbpe}-2b/v1` (promoted under the eduLLM dataset standard).
- **Matching:** equal tokens—$\approx$ 2.0B train tokens total, $6 \times 333{,}316{,}096$ per language—not equal bytes.
- **Optimization:** 7629 steps, sequence length 2048, global batch 262,144 tokens ($7629 \times 262{,}144 = 1{,}999{,}896{,}576$ tokens), warmup 200 steps, checkpoint every 2000 steps; final checkpoint `step7629`.
- **Compute:** 8×A10G (`gpu-8xa10g`), team `scratch`, W&B project `plan-b-phase0`.

Evaluation specifications. Held-out `val` shards only (no train-shard leakage). We load `step7629` with OLMo-core `LMEvaluator` under padded FSL; Phase 0 val shards contain no EOS, so instances are non-overlapping sequence-length windows over each shard (final chunk may be shorter and pad-masked). Primary metric is bits/token = CE/ ln 2. Because SuperBPE tokens cover more text on average, we also report an approximate bits-per-byte

$$\widetilde{\mathrm{BPB}}(\ell) = \frac{\mathrm{bits/token}(\ell)}{\mathrm{chars/token}(\ell)}, \tag{3}$$

where chars/token comes from the Study B FLORES-200 `devtest` suite (non-whitespace Unicode code points over CTC). This tracks UTF-8 BPB closely on Latin scripts; absolute levels on Hindi and Mandarin are inflated relative to true UTF-8 bytes, but within-language BPE-versus-SuperBPE rankings remain informative.

Results. Table 6 shows raw bits/token. Micro-average bits/token is 5.45 (BPE) versus 6.29 (SuperBPE), a gap of +0.84 favoring BPE; Mandarin is essentially tied (−0.05). Table 7 shows that fertility normalization collapses most of that gap: non-CJK micro $\widetilde{\mathrm{BPB}}$ differs by only $\approx +0.04$, and SuperBPE is slightly better on Haitian Creole alone. Equal-token matching itself biases against SuperBPE (fewer unique bytes seen for the same token budget), so Phase 0 is a smoke gate, not a kill gate for the equal-byte 1B comparison.

6.3 Phase 1: 1B UniMax bake-off (deferred)

We plan a matched OLMo-class [OLMo Team et al., 2025] ∼1B pretraining run for each arm under the protocol that actually matches the SuperBPE hypothesis:

Table 6: Phase 0 held-out val LM eval (`olmo2_100M` @ `step7629`, equal-token $\sim$2B). Bits/token $= \text{CE}/\ln 2$.

Language	BPE bits	BPE PPL	SuperBPE bits	SuperBPE PPL	Δ bits
English	5.568	47.43	6.716	105.10	+1.148
Haitian Creole	5.913	60.26	7.105	137.66	+1.192
Hindi	4.270	19.29	5.270	38.59	+1.000
Hungarian	5.132	35.07	5.993	63.71	+0.861
Swahili	5.316	39.84	6.175	72.25	+0.859
Mandarin	6.521	91.85	6.476	88.99	−0.046
Micro	5.453	43.82	6.289	78.20	+0.836

Table 7: Phase 0 approximate BPB: bits/token divided by Study B FLORES chars/token. Negative Δ favors SuperBPE. Absolute Mandarin/Hindi levels are inflated vs. true UTF-8 BPB.

Language	BPE $\widetilde{\text{BPB}}$	SuperBPE $\widetilde{\text{BPB}}$	Δ
English	1.450	1.488	+0.038
Haitian Creole	1.785	1.772	−0.013
Hindi	1.459	1.529	+0.070
Hungarian	1.332	1.415	+0.084
Swahili	1.346	1.363	+0.016
Mandarin	4.078	4.086	+0.008
Micro (non-CJK)	1.475	1.514	+0.039

- **Model:** `olmo2_1B`; sequence length 2048; BPE arm 80,200 steps (≈21B tokens); SuperBPE continues from equal-bytes (≈71,986 steps) to equal FLOPs at 80,200.
- **Corpora:** `pretrain/fineweb2-unimax-{bpe,superbpe}-20b/v1`.
- **Matching:** same *bytes* (not the same token count); SuperBPE then catches up to equal FLOPs.
- **Mixture:** UniMax [Chung et al., 2023] with a 4-epoch cap [Muennighoff et al., 2023]; Haitian Creole is the binding language (≈ 3.96%), the other five ≈ 19.21% each.
- **Compute plan:** `gpu-8xh100` (≈$1.8k both arms at equal FLOPs).
- **Eval:** true UTF-8 bits-per-byte on held-out val / FLORES (primary); AmericasNLP placeholder unchanged.

Tables 8 and 9 are placeholders to be filled after training.

7 Discussion

Replacing the 3–12× range. Study A shows that the familiar range is not wrong so much as underspecified. On `o200k`, Amharic (5.78×) and Odia (4.99×) sit inside it; on Llama/GLM, Odia exceeds 12×. Dialectal Arabic and the Swahili/Hausa controls show that African languages are not uniformly taxed: Latin-script Swahili/Hausa are near 1.5–2×, while Ethiopic Amharic is several times worse. Reporting language × tokenizer cells is more actionable for API cost planning than a single interval.

Table 8: Deferred: bits-per-byte on FLORES-200 after ∼1B OLMo pretraining (20B-token BPE budget). *To be filled after Plan B 1B training.*

Language	BPE BPB	SuperBPE BPB
English	*TBD*	*TBD*
Hungarian	*TBD*	*TBD*
Mandarin	*TBD*	*TBD*
Hindi	*TBD*	*TBD*
Swahili	*TBD*	*TBD*
Haitian Creole	*TBD*	*TBD*
Mean	*TBD*	*TBD*

Table 9: Deferred: bits-per-byte on AmericasNLP evaluation sets. *To be filled after Plan B 1B training.*

Language / set	BPE BPB	SuperBPE BPB
(language)	*TBD*	*TBD*
(language)	*TBD*	*TBD*
Mean	*TBD*	*TBD*

Controls matter. Without Swahili and Hausa, one might attribute the Africa continent mean (2.94 on `o200k`) entirely to "African languages." The control row shows that the Amharic outlier drives much of that mean; Latin-script African languages behave closer to other Latin mid-resource languages.

SuperBPE is language-conditional. Study B's Mandarin and Hungarian failures on English-relative premium are the load-bearing result for tokenizer choice, and they corroborate the BPB ranking of Chowdhury and Woolf [2026] on the same two languages. Fertility alone would recommend SuperBPE for Hungarian; premium—the API-relevant parallel-length ratio—would not.

Phase 0 does not settle 1B. Raw bits/token in Table 6 would appear to kill SuperBPE; fertility-normalized Table 7 shows most of that gap was token length, not modeling quality, and the ranking remains language-conditional (Haitian Creole favors SuperBPE; Hungarian and Hindi favor BPE). Phase 0 also used equal-*token* matching, which starves SuperBPE of bytes relative to Plan B's equal-byte protocol. Until Tables 8–9 are filled under equal bytes and true UTF-8 BPB, we treat encode-time premium as the conservative decision metric for cost-sensitive deployments, with Chowdhury and Woolf's extrinsic gap as prior evidence that LM quality is unlikely to reverse the Mandarin/Hungarian call.

8 Limitations

Study A is encode-only: it measures token tax, not downstream task quality. Study B's language set excludes Amharic, Odia, Hausa, and dialectal Arabic, so the SuperBPE recommendation does not yet cover Study A's highest-tax scripts. The three studies in this codebase (12-language efficiency, 18-language Zipf, 6-language Plan A/B) use non-overlapping language locks; numbers must not be pooled without labels. Gigatoken / `supergigatoken`'s Hindi SuperBPE stage-2 deviation (§5.3) means Hindi SuperBPE premiums are not directly comparable to published Liu et al. numbers. Phase 0 matches equal tokens rather than equal bytes, and its $\widetilde{\text{BPB}}$ uses FLORES characters-per-token rather than UTF-8 byte lengths on the Phase 0 val shards; Latin scripts approximate true BPB, while Hindi and Mandarin absolute levels are inflated. The 1B LM bake-off is specified but not executed; Tables 8–9 are empty by design.

9 Conclusion

We measured a FLORES token-tax table for Amharic, Odia, and dialectal Arabic with Swahili/Hausa controls. Tokenizer-specific premiums replace the literature's 3–12× range: Amharic 5.78× and Odia 4.99× on `o200k`; Odia above 12× on Llama and GLM. Because APIs bill per token, these premiums are relative cost multipliers for parallel content. A matched SuperBPE-versus-BPE bake-off at equal-byte scale shows SuperBPE improving fertility broadly but worsening English-relative premium for Mandarin and Hungarian, so tokenizer choice must be language-conditional. A Phase 0 `olmo2_100M` pilot shows that raw bits/token strongly favors BPE under equal-token training, but fertility-normalized approximate BPB largely closes the gap and remains language-conditional. Filling the deferred 1B equal-byte BPB tables is the remaining step before a quality-aware recommendation.

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FLORES-200 [Goyal et al., 2022, NLLB Team et al., 2022], FineWeb /FineWeb-2 [Penedo et al., 2024, 2025], SuperBPE [Liu et al., 2025], Gigatoken [Rød, 2026], `supergigatoken` [Verma, 2026], and the OLMo stack [OLMo Team et al., 2025] made these measurements possible.

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Table 10: Full English-relative token premium matrix (Study A, FLORES-200 `devtest`).

Language	o200k	Llama	Qwen	GLM	SuperBPE
English (`eng_Latn`)	1.00	1.00	1.00	1.00	1.00
Hungarian (`hun_Latn`)	1.79	2.12	2.12	1.89	2.92
Mandarin (`zho_Hans`)	1.25	1.28	1.00	0.95	2.19
Ukrainian (`ukr_Cyrl`)	1.75	1.64	2.50	1.83	3.84
Swahili (`swh_Latn`)	1.49	1.93	1.94	1.93	2.43
Hausa (`hau_Latn`)	1.55	1.97	1.94	1.94	2.47
Amharic (`amh_Ethi`)	5.78	7.59	4.05	7.59	5.90
Odia (`ory_Orya`)	4.99	12.16	9.70	12.36	13.53
Egyptian Arabic (`arz_Arab`)	1.41	1.65	1.63	1.75	3.22
Moroccan Arabic (`ary_Arab`)	1.48	1.71	1.69	1.81	3.26
Quechua (`quy_Latn`)	1.83	2.07	2.06	2.04	2.54
Guarani (`grn_Latn`)	1.78	2.08	2.09	2.04	2.77

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A Full Study A Premium Matrix

B Additional Study B Panels

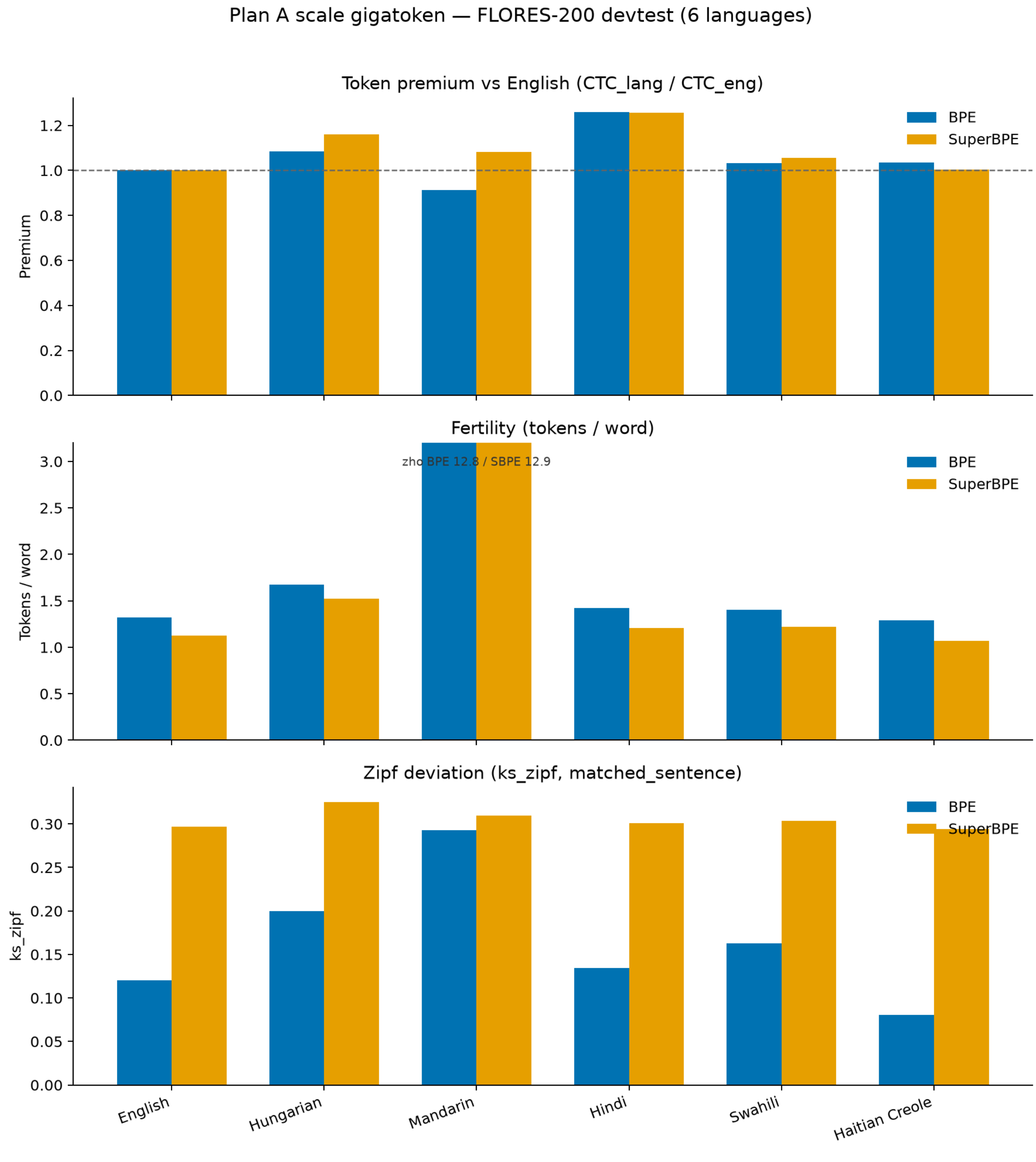

Figure 2: Study B FLORES suite summary: token premium, fertility, and ks_zipf for matched BPE and SuperBPE.

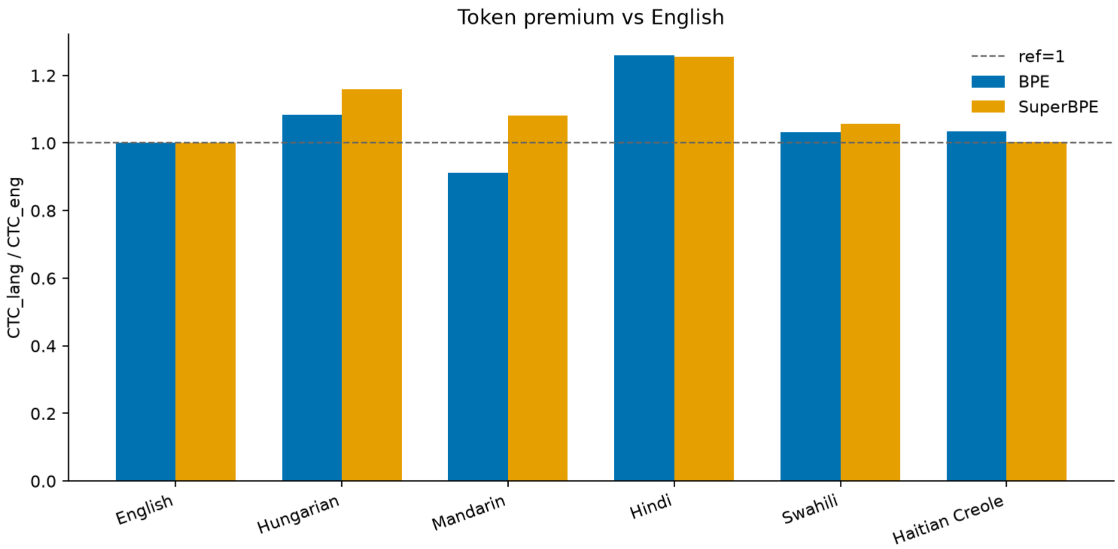

Figure 3: Study B token premium by language.

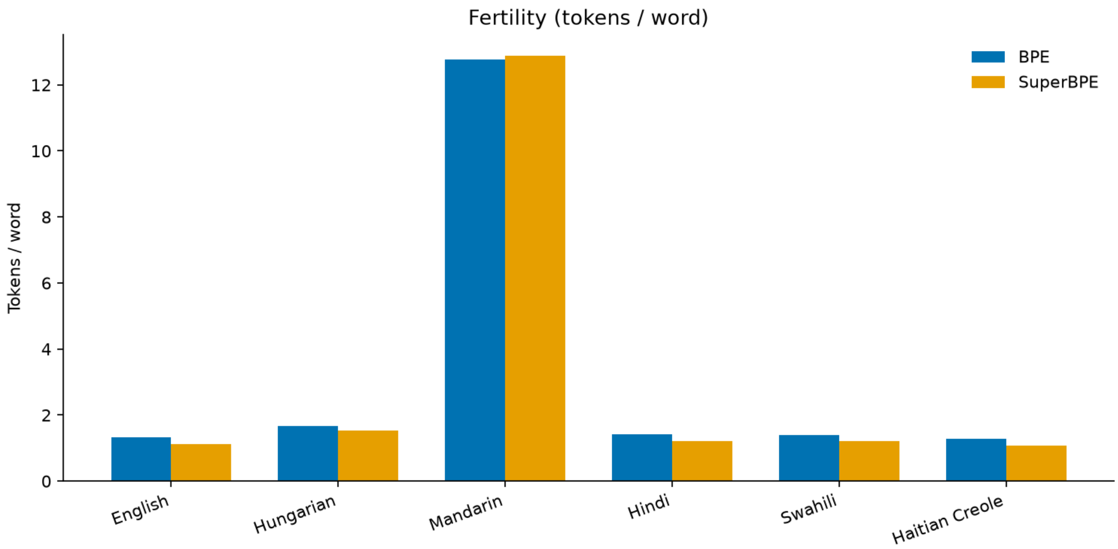

Figure 4: Study B fertility by language.

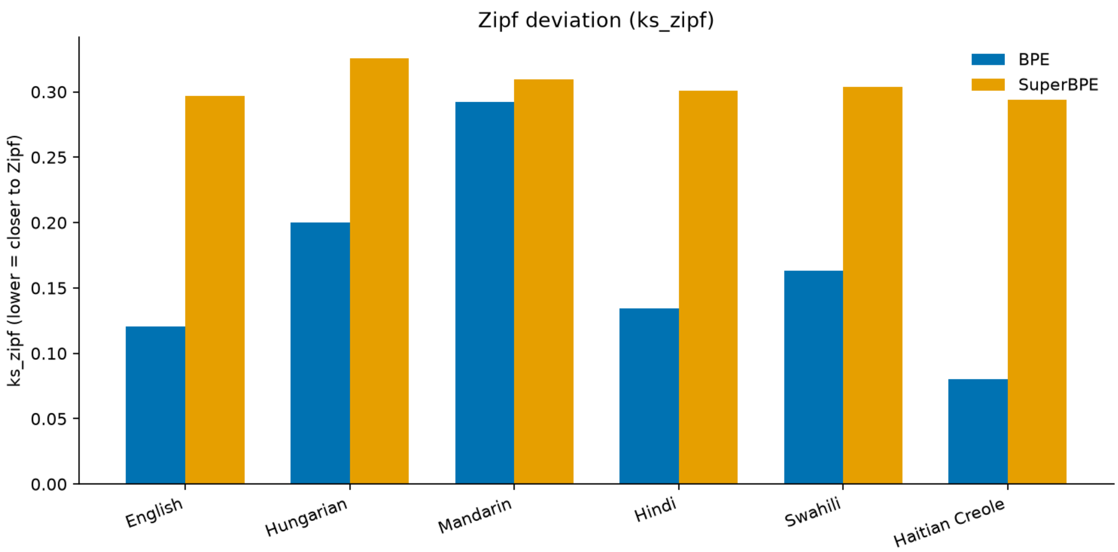

Figure 5: Study B ks_zipf by language.